

Split ring resonator biosensor-an innovative design and analysis

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II. PROPOSED BIOSENSOR

In this paper single ring double split SRR biosensor with its dimensions is shown in Fig 1.

Abstract—In this study, a split ring resonator (SRR) design is simulated for the detection of different cancer cells in biosensing application using COMSOL Multiphysics software. Finite element method (FEM) is used to evaluate electric field distribution in SRR. The analysis of biosensor is done in microwave region. Small change in dielectric constant produces shift in resonant peak of SRR so this biosensor can be used for the detection of different types of cancer cells like Hela, PC12, MDA-MB-231, MCF-7, Jurkat. The advantages of this biosensor are low cost manufacturing and level free sensing with appropriate sensitivity.

Index Terms—Split ring resonators; Finite element method; cancer cell detection; Biosensors.

I. INTRODUCTION

Currently, the research in the field of biosensor is popular in the area of physics, environment, medicine etc. The pH estimation, DNA investigation, cancer detection and molecular concentrations are characterized by the biosensors [1]. Metamaterials are defined as structure materials which have anomalous values of permittivity and permeability. These materials have negative values of permeability and permittivity [2], [3] and they exhibit many applications like biosensing, notch filters etc at different frequency range [4]–[9], [14]–[16]. In recent years using split ring resonator researchers have done early detection of cancer cell [5], detection of biotin and streptavidin binding [6], prostate-specific antigen [7], immune globulin [8], and some observed DNA hybridization [9] in biosensing application.

SRR is a pair of concentric loops with a dielectric substrate. A split in them present at opposite ends, which helps metamaterials to produce the required magnetic susceptibility. Because of its size, shape and materials every SRR has its own resonance frequency for the electromagnetic field [10]–[13]. SRR is very sensitive to change in dielectric constant, and thus change in dielectric constant of different samples provides shift in resonance frequency of SRR [4]–[9], [14]–[16].

In this paper we proposed SRR biosensor with a double split single ring which reduces the cost of SRR with good sensitivity for the detection of different cancer cells. COMSOL multiphysics is used to simulate the proposed biosensor. Researcher also fabricated such type of biosensor [17], [18].

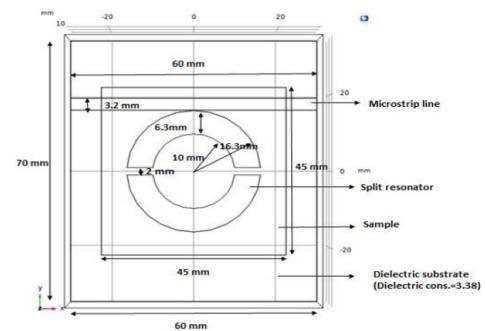


Fig. 1. Schematic diagram of Single ring Double Split SRR Biosensor.

Using proposed biosensor different cancer cells are detected whose dielectric constant are given in Table I [17].

TABLE I
DIELECTRIC CONSTANT OF NORMAL CELL AND DIFFERENT CANCER CELL

Cancer Cell	Dielectric Constant
Normal Cell	1.822500
Hela	1.937660
PC12	1.946025
MDA-MB-231	1.957201
MCF-7	1.962801
Jurkat	1.932100

III. SIMULATION AND RESULTS

The biosensor is designed and simulated in COMSOL Multiphysics software. FEM method is used to solve the electromagnetic field equation. In simulation, we used physics controlled mesh with normal element size. For biosensor analysis the transmission coefficient (S21) of port 2 with respect to port 1 is evaluated for different cancer cells and compared with normal cell transmission coefficient as shown in Figure 2.

It can be seen the minimum transmission coefficient is observed at 2.3042 GHz for the normal cell and the shift in this

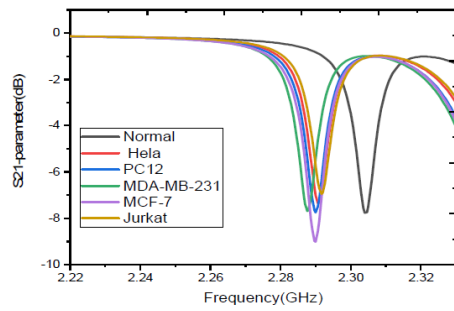


Fig. 2. Transmittance of normal and different cancer cells in frequency range 2.22GHz to 2.33 GHz.

TABLE II
RESONANCE AND SENSITIVITY OF DIFFERENT CANCER CELL

Cancer cell	Resonance frequency (GHz)	Sensitivity (GHz)
Hela	2.2908	0.0134
PC12	2.2900	0.0142
MDA-MB-231	2.2876	0.0166
MCF-7	2.2898	0.0144
Jurkat	2.2916	0.0126

resonance frequency for different cancer cells are calculated and shown in Table II.

The electromagnetic field distribution for different cancer cells are shown in Figure 3. The sensitivities calculated for this biosensor for different cancer cell are comparable with SRR biosensor defined in reference [15], where as we use only one split ring so the proposed biosensor is cost effective.

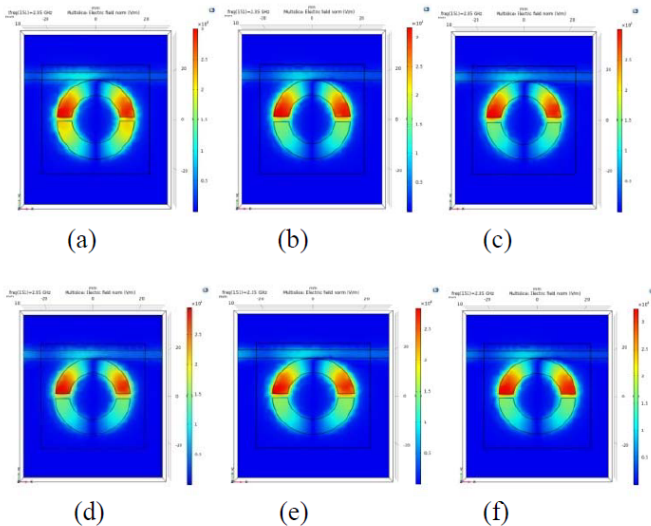


Fig. 3. Electric field distribution of (a) Normal cell (b) Hela cell (c) PC12 cell (d) MDA-MB-231 cell (e) MCF-7cell (f) Jurkat cell.

IV. CONCLUSIONS

In this paper, we simulated double split single ring resonator biosensor by COMSOL multiphysics software. Good

sensitivity for different cancer cells are achieved with only double split single ring resonator which make the proposed SRR sensor cost effective so it may be used for different cancer cell detection in biosensing application.

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